

June 2026

To the Editors,
Patterns, Cell Press
Attn: Dr. Alejandra Alvarado

Dear Dr. Alvarado,

Thank you for your considered assessment of our earlier submission (PATTERNS-D-26-00307) and for the invitation to submit a revised and expanded version. We are resubmitting as a new manuscript that directly implements your central recommendation: to **consolidate the OmniGene-4 and BioPAWS work into a single, balanced study with a fair benchmarking framework**. We summarize below exactly how this version addresses the concern you raised.

Your concern. You wrote that you were not fully convinced the paper presented a sufficiently fair benchmarking framework, “particularly given that BioPAWS was also developed by your group,” and you suggested an integrated manuscript consolidating the two efforts. We agree this was the right diagnosis: in the previous version, the central model was evaluated largely on a benchmark from the same group that was neither independently described nor released as a standing resource, which understandably reads as self-benchmarking.

What we changed. We have turned the benchmark itself into a first-class, openly released contribution and re-positioned OmniGene-4-MM as *one of several models evaluated on it*, rather than its author-evaluator. Concretely:

1. **BioPAWS-2, a public benchmark and training resource (new Section 2).** We consolidate and substantially expand the benchmark into **BioPAWS-2**: a chat-form instruction-tuning suite spanning **nine task families and 22 tasks (~306K examples)** — pairwise homology, protein function, DNA/genomic classification, variant effect, structure, cross-modal DNA $\leftrightarrow$ protein, chain-of-thought reasoning, biomedical QA, and multimodal image+text. It is released openly on Hugging Face ([dnagpt/biopaws-2](#)) with all conversion and evaluation code on GitHub, so that *any* group can train and score *any* model on it.
2. **A fair, paradigm-neutral evaluation protocol.** Every task is scored under two modes: zero-shot QA (for generative LLMs) and fine-tune-then-evaluate, where a specialized protein/DNA language model runs its *native* frozen-backbone head-training protocol and a generalist LLM runs LoRA-SFT — on the same train/test split, each in its home-turf regime. This lets a specialist PLM (e.g. ESM-2-3B) and a generalist LLM be compared on a common axis without disadvantaging either.
3. **OmniGene-4-MM is one entrant among several.** The leaderboard reports OmniGene-4-MM alongside ESM-2-3B (per-task heads) and Qwen-3.7-max (zero-shot), across all families. Where the specialist is stronger (structure, localization) we report it plainly; the contribution is that a single generalist covers all tasks with one training run while remaining competitive, not that our model wins every cell.
4. **New, independent findings that strengthen rather than assume the result.** Consolidation surfaced results we could not have shown in the narrower framing: (a) on ProteinGym

variant-effect prediction, a mean-pooled PLM head sits at chance (0.53) because single-point-mutation signal is averaged away, whereas the chat formulation — presenting the mutation as explicit tokens — recovers it (0.67 after SFT); and (b) fine-grained biological label sets (protein fold, localization) are near-zero zero-shot for *every* large model tested, quantifying precisely what the BioPAWS-2 training corpus contributes.

Reframing and data integrity. Reflecting where the evidence now points, we have retitled and re-led the manuscript around **BioPAWS-2 as the central contribution**, with OmniGene-4 presented as a case study evaluated on it. Two additions make the benchmark self-auditing. First, a matched *raw-Gemma-base* control under the identical joint fine-tuning protocol shows that our model’s biological continued-pretraining confers *no material advantage under supervised fine-tuning* (macro 0.769 vs. 0.760) — i.e. we used our own benchmark to argue *against* a self-serving conclusion, which is the opposite of self-benchmarking. Second, we release an MMseqs2 leakage audit and re-split every sequence task *entity-disjoint* (homology leakage reduced from 0.78 to 0.08); re-running all supervised results on the leakage-free splits moves the headline by only -1.4 points, demonstrating the leaderboard is not a memorization artifact. ProteinGym is additionally reported with its field-standard Spearman metric.

What is unchanged. The empirical core of the original submission — the router-level CPT-vs-SFT decomposition, the remote-homology results, and the modality-invariant-transfer analysis of OmniGene-4-MM — is retained in full (Sections 3–5). The new BioPAWS-2 section provides the independent benchmarking foundation those results now rest on.

Openness. All artefacts are public: the BioPAWS-2 dataset ([dnagpt/biopaws-2](https://github.com/dnagpt/biopaws-2)), the model checkpoints ([dnagpt/OmniGene-4-MM-merged](https://github.com/dnagpt/OmniGene-4-MM-merged) and the LoRA adapter), and all training, conversion, evaluation, leakage-audit, and leaderboard code (github.com/maris205/omnigene4). A third party can reproduce every leaderboard number — including the leakage audit and the entity-disjoint splits — or add a new model, without any artefact private to our group.

Preprint and prior consideration. A preliminary version is on bioRxiv (DOI [10.1101/2026.01.03.697478](https://doi.org/10.1101/2026.01.03.697478)); an updated v2 with the multi-modal extension is at DOI [10.64898/2026.05.12.724542](https://doi.org/10.64898/2026.05.12.724542). The present manuscript adds the consolidated BioPAWS-2 benchmark and its independent evaluation. The author declares no competing interests; GPU compute was donated by io.net, which exercised no influence over the research.

We are grateful for the path you offered toward a more compelling and balanced paper, and we believe this integrated version answers it. We welcome any further feedback.

Sincerely,

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